

# Offline-first sync without tears

Maya Okafor · Tidewater Labs · Harbor Dev Summit 2026

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## TL;DR

- Apps write locally first and reconcile with the server as a peer, without waiting on it.
- Field-level merges and delete tombstones help preserve edits and prevent stale devices from restoring deleted data.
- Tests should simulate airplane mode, five minutes of clock skew, and app termination mid-write.

## Overview

Maya Okafor presented offline-first reliability as a question of preserving user work when connectivity and execution are unreliable. The approach keeps local interactions available, then reconciles data through explicit conflict rules. Testing targets failures in connectivity, clocks, and writes rather than assuming a stable online session.

## Local-first writes

Because the network may be unavailable, the app writes locally on every action rather than waiting for connectivity.

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## The network is a rumour

- Write locally first, always
- Treat the server as a peer, not the truth
- Show sync state, never block on it

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### Server reconciliation

After local writes, the app reconciles with the server as a peer, not as the source of truth, and shows sync state without blocking the user.

## Conflict handling

Conflict resolution affects whether users keep their work: last-writer-wins can discard edits, while field-level merging preserves changes at a finer level. Tombstones for deletes prevent stale devices from restoring removed data.

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# Conflicts are product decisions

- Last-writer-wins loses user work
- Merge fields, not records
- Keep a tombstone for every delete

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## Failure testing

Tests should exercise airplane mode, five minutes of clock skew in either direction, and app termination during a write.

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# Test it on a train

- Airplane mode in CI
- Clock skew of  $\pm 5$  minutes
- Kill the app mid-write

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